**INTERNATIONAL SEARCH REPORT**

International application No.
**PCT/CA2021/051866**

A. CLASSIFICATION OF SUBJECT MATTER

IPC: ***A61B 5/11*** (2006.01) , ***G16H 50/30*** (2018.01)

CPC: , A61B 5/11 (2020.01) , A61B 5/1122 (2020.01) , G16H 50/30 (2020.01)

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
IPC: A61B 5/11 (2006.01), G16H 50/30 (2018.01)

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic database(s) consulted during the international search (name of database(s) and, where practicable, search terms used)
Database: Questel Orbit.

Keywords: Biomechanic+, Risk data, Assessment

C. DOCUMENTS CONSIDERED TO BE RELEVANT

| Category* | Citation of document, with indication, where appropriate, of the relevant passages | Relevant to claim No. |
|---|---|---|
| X | US 20180177436 A1 (CHANG, A.R., et al.) 28 June 2018 (28-06-2018)<br>* Abstract; [0024], [0028], [0032] , [0065], [0068], [0070], [0071], [0104], [0105], [0160]; Figs. 1-3, 7, 8 * | 1-18 |
| A | WO 2019213399 A1 (NAJAFI, B. et al.) 7 November 2019 (07-11-2019)<br>* Entire document * | 1-18 |
| A | JP 2020120807 A (HIMENO, R. et al.) 13 August 2020 (13-08-2020)<br>* Entire document * | 1-18 |

☐ Further documents are listed in the continuation of Box C. ☐ See patent family annex.

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|---|---|
| * | Special categories of cited documents: |
| "A" | document defining the general state of the art which is not considered to be of particular relevance |
| "D" | document cited by the applicant in the international application |
| "E" | earlier application or patent but published on or after the international filing date |
| "L" | document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified) |
| "O" | document referring to an oral disclosure, use, exhibition or other means |
| "P" | document published prior to the international filing date but later than the priority date claimed |
| "T" | later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention |
| "X" | document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone |
| "Y" | document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art |
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|---|---|
| Date of the actual completion of the international search<br>01 March 2022 (01-03-2022) | Date of mailing of the international search report<br>20 April 2022 (20-04-2022) |
| Name and mailing address of the ISA/CA<br>Canadian Intellectual Property Office<br>Place du Portage I, C114 - 1st Floor, Box PCT<br>50 Victoria Street<br>Gatineau, Quebec K1A 0C9<br>Facsimile No.: 819-953-2476 | Authorized officer<br>Andy Wong (819) 639-8351 |

C (Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

| Category* | Citation of document, with indication, where appropriate, of the relevant passages | Relevant to claim No. |
|---|---|---|
| A | KR 10-1974831 B1 (XIONG, S. et al.) 3 May 2019 (03-05-2019)<br>* Entire document * | 1-18 |
| E, X | WO 2022026658 A1 (ZELIK, K. et al.) 3 February 2022 (03-02-2022)<br>* Entire document * | 1-18 |

**INTERNATIONAL SEARCH REPORT**
Information on patent family members

International application No.
**PCT/CA2021/051866**

| Patent Document Cited in Search Report | Publication Date | Patent Family Member(s) | Publication Date |
|---|---|---|---|
| US2018177436A1 | 28 June 2018 (28-06-2018) | None | |
| WO2019213399A1 | 07 November 2019 (07-11-2019) | EP3787502A1<br>US2021236021A1 | 10 March 2021 (10-03-2021)<br>05 August 2021 (05-08-2021) |
| JP2020120807A | 13 August 2020 (13-08-2020) | None | |
| KR101974831B1 | 03 May 2019 (03-05-2019) | KR20180067233A | 20 June 2018 (20-06-2018) |
| WO2022026658A1 | 03 February 2022 (03-02-2022) | None | |